

ISyE 6525 HW3 — Question 1

Q1(a) Multilinear algebra

Given

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

(i) Kronecker product $A \otimes B$

Using the block definition,

$$A \otimes B = \begin{bmatrix} 1B & 2B \\ 3B & 4B \end{bmatrix} = \begin{bmatrix} 5 & 6 & 10 & 12 \\ 7 & 8 & 14 & 16 \\ 15 & 18 & 20 & 24 \\ 21 & 24 & 28 & 32 \end{bmatrix}.$$

(ii) Khatri-Rao product $A \odot B$

The Khatri-Rao product is the matching-column Kronecker product:

$$A \odot B = \begin{bmatrix} a_1 \otimes b_1 & a_2 \otimes b_2 \end{bmatrix}.$$

Thus,

$$a_1 \otimes b_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \otimes \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 15 \\ 21 \end{bmatrix},$$

and

$$a_2 \otimes b_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \otimes \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 12 \\ 16 \\ 24 \\ 32 \end{bmatrix}.$$

Therefore,

$$A \odot B = \begin{bmatrix} 5 & 12 \\ 7 & 16 \\ 15 & 24 \\ 21 & 32 \end{bmatrix}.$$

(iii) Hadamard product $A * B$

Elementwise multiplication gives

$$A * B = \begin{bmatrix} 1 \cdot 5 & 2 \cdot 6 \\ 3 \cdot 7 & 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 5 & 12 \\ 21 & 32 \end{bmatrix}.$$

(iv) $(A \otimes B)^{-1}$

Starting from

$$\left[\begin{array}{cccc|cccc} 5 & 6 & 10 & 12 & 1 & 0 & 0 & 0 \\ 7 & 8 & 14 & 16 & 0 & 1 & 0 & 0 \\ 15 & 18 & 20 & 24 & 0 & 0 & 1 & 0 \\ 21 & 24 & 28 & 32 & 0 & 0 & 0 & 1 \end{array} \right],$$

row reduction gives the upper-triangular form

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -12 & -16 & -4 & 0 & 0 & 1 \\ 0 & 8 & 98 & 128 & 28 & 1 & 0 & -7 \\ 0 & 0 & 10 & 12 & 3 & 0 & -1 & 0 \\ 0 & 0 & 0 & 4 & 21 & -15 & -7 & 5 \end{array} \right].$$

Now eliminate the fourth column:

$$R_1 \leftarrow R_1 + 4R_4, \quad R_3 \leftarrow R_3 - 3R_4, \quad R_2 \leftarrow R_2 - 32R_4,$$

which gives

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -12 & 0 & 80 & -60 & -28 & 21 \\ 0 & 8 & 98 & 0 & -644 & 481 & 224 & -167 \\ 0 & 0 & 10 & 0 & -60 & 45 & 20 & -15 \\ 0 & 0 & 0 & 4 & 21 & -15 & -7 & 5 \end{array} \right].$$

Normalize the last two pivots:

$$R_3 \leftarrow \frac{1}{10}R_3, \quad R_4 \leftarrow \frac{1}{4}R_4.$$

Then

$$R_2 \leftarrow R_2 - 98R_3, \quad R_2 \leftarrow \frac{1}{8}R_2, \quad R_1 \leftarrow R_1 + 12R_3.$$

Therefore,

$$(A \otimes B)^{-1} = \begin{bmatrix} 8 & -6 & -4 & 3 \\ -7 & 5 & \frac{7}{2} & -\frac{5}{2} \\ -6 & \frac{9}{2} & 2 & -\frac{3}{2} \\ \frac{21}{4} & -\frac{15}{4} & -\frac{7}{4} & \frac{5}{4} \end{bmatrix}.$$

(v) $A^{-1} \otimes B^{-1}$

First,

$$A^{-1} = \frac{1}{1 \cdot 4 - 2 \cdot 3} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix},$$

and

$$B^{-1} = \frac{1}{5 \cdot 8 - 6 \cdot 7} \begin{bmatrix} 8 & -6 \\ -7 & 5 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ \frac{7}{2} & -\frac{5}{2} \end{bmatrix}.$$

Hence

$$A^{-1} \otimes B^{-1} = \begin{bmatrix} 8 & -6 & -4 & 3 \\ -7 & 5 & \frac{7}{2} & -\frac{5}{2} \\ -6 & \frac{9}{2} & 2 & -\frac{3}{2} \\ \frac{21}{4} & -\frac{15}{4} & -\frac{7}{4} & \frac{5}{4} \end{bmatrix}.$$

Thus,

$$\boxed{(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}}.$$

Q1(b)

Let

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{r \times s}, \quad C \in \mathbb{R}^{n \times p}, \quad D \in \mathbb{R}^{s \times t}.$$

We want to show

$$(A \otimes B)(C \otimes D) = AC \otimes BD.$$

Using the block representation,

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix},$$

and

$$C \otimes D = \begin{bmatrix} c_{11}D & \cdots & c_{1p}D \\ c_{21}D & \cdots & c_{2p}D \\ \vdots & \ddots & \vdots \\ c_{n1}D & \cdots & c_{np}D \end{bmatrix}.$$

The (i, j) -th block of the product is

$$\begin{aligned} [(A \otimes B)(C \otimes D)]_{ij} &= \sum_{k=1}^n (a_{ik}B)(c_{kj}D) \\ &= \sum_{k=1}^n a_{ik}c_{kj}BD \\ &= \left(\sum_{k=1}^n a_{ik}c_{kj} \right) BD \\ &= (AC)_{ij}BD. \end{aligned}$$

This is exactly the (i, j) -th block of $AC \otimes BD$. Hence

$$\boxed{(A \otimes B)(C \otimes D) = AC \otimes BD}.$$

The dimensions also agree:

$$A \otimes B \in \mathbb{R}^{mr \times ns}, \quad C \otimes D \in \mathbb{R}^{ns \times pt},$$

so the product lies in $\mathbb{R}^{mr \times pt}$.

Q1(c)

Let

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{p \times q}, \quad X \in \mathbb{R}^{q \times n}, \quad Y \in \mathbb{R}^{p \times m}.$$

We want to show

$$\text{vec}(Y)^\top (A \otimes B) \text{vec}(X) = \text{tr}(A^\top Y^\top B X).$$

Use the identities

$$\text{vec}(AXB) = (B^\top \otimes A) \text{vec}(X),$$

and

$$\text{vec}(U)^\top \text{vec}(V) = \text{tr}(U^\top V).$$

Choosing the left matrix as B and the right matrix as $A^\top$,

$$(A \otimes B) \text{vec}(X) = \text{vec}(BXA^\top).$$

Therefore,

$$\begin{aligned} \text{vec}(Y)^\top (A \otimes B) \text{vec}(X) &= \text{vec}(Y)^\top \text{vec}(BXA^\top) \\ &= \text{tr}(Y^\top BXA^\top). \end{aligned}$$

Using the cyclic property of the trace,

$$\text{tr}(Y^\top BXA^\top) = \text{tr}(A^\top Y^\top BX).$$

Hence

$$\boxed{\text{vec}(Y)^\top (A \otimes B) \text{vec}(X) = \text{tr}(A^\top Y^\top BX)}.$$